

# Geography of Colombian Life Satisfaction

Monday 27<sup>th</sup> April, 2026 19:44

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This paper is about geography of Colombian happiness.<sup>1</sup> We start with departments/provinces, then move to cities/municipalities, and finally finish up with urban-rural and farmers.

about colombia boilerplate like 2nd biggest land mas and pop in lat am? one of the happiest in the world:

**Table 1:** 10 happiest countries in the world. Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at [worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life](http://worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life); and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at <https://www.worldvaluessurvey.org>. Technical information for WDH is at <https://worlddatabaseofhappiness.eur.nl/reports/finding-reports-on-happiness-in-nations/technical-details-to-rank-reports-of-happiness-in-nations/technical-details-to-rank-report-average-happiness-in-nations-2010-2019> and WVS documentation is at <https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp>.

| WDH  |             |                  | WVS  |             |                  |
|------|-------------|------------------|------|-------------|------------------|
| rank | country     | happiness (1-10) | rank | country     | happiness (1-10) |
| 1    | Denmark     | 8.2              | 1    | Puerto Rico | 8.4              |
| 2    | Mexico      | 8.1              | 2    | Mexico      | 8.3              |
| 3    | Colombia    | 8.1              | 3    | Colombia    | 8.3              |
| 4    | Switzerland | 8                | 4    | Qatar       | 8.0              |
| 5    | Finland     | 8                | 5    | Norway      | 7.9              |
| 6    | Iceland     | 8                | 6    | Nicaragua   | 7.9              |
| 7    | Costa Rica  | 7.9              | 7    | Tajikistan  | 7.9              |
| 8    | Norway      | 7.9              | 8    | Switzerland | 7.9              |
| 9    | Canada      | 7.9              | 9    | Uzbekistan  | 7.9              |
| 10   | Qatar       | 7.8              | 10   | Ecuador     | 7.8              |

and need about geography like goog 'gepgraphy of colombia' in goog scholar and normal goog

about roads etc ranked one of thw worst and airports

We know little about colombian happiness: my paper, stiupid burger as cited in my paper, and that govt report i cite in my paper!! do skim thru it!!

burger few observations: only few thousand (7-9k depending on model) and problematic measurement across space as elaborated in urb unhappiness is common

and there is interesting (Peiró-Palomino et al. 2020) but it uses an index objective indicators of quality of life such as income, housing and pollution.

There is a recent United Nations study on Colombian happiness using the same Encuesta Nacional de Calidad de Vida as we use, just earlier waves (PNUD 2023). The study is available from the the UN<sup>2</sup>

<sup>1</sup>Following usual practice in social indicators research terms happiness and life satisfaction are used interchangeably, and specifically we mean cognitive and evaluative life satisfaction rather than affective happiness.

<sup>2</sup>[https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp\\_co\\_pub\\_Percepciones\\_bienestar\\_subjetivo\\_2do\\_cuaderno\\_INDH\\_Feb21\\_2023.pdf](https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp_co_pub_Percepciones_bienestar_subjetivo_2do_cuaderno_INDH_Feb21_2023.pdf).

It is closest to our study and most useful overview of Colombian happiness geography to date. There is in fact a similar map to ours in Figure 1.9 on page 26 for 2018-2021, but it is difficult to read as the colormap is undistinctive. While happy cafetero agrees with our analysis, happy Narino does not. Likewise it finds Guaviare to be unhappy, while we find it moderately happy. A possible explanation for the differences is different timeframe: 2018-2021 theirs v 2023 and 2021 ours. Likewise in figure 1.10 they break down SWB by urban-rural, ours agrees that there is almost no difference, and we add farmer and by income.

## 1 Data

We use a great source of data ECV. We use 2023 edition.

The life satisfaction item reads: ??

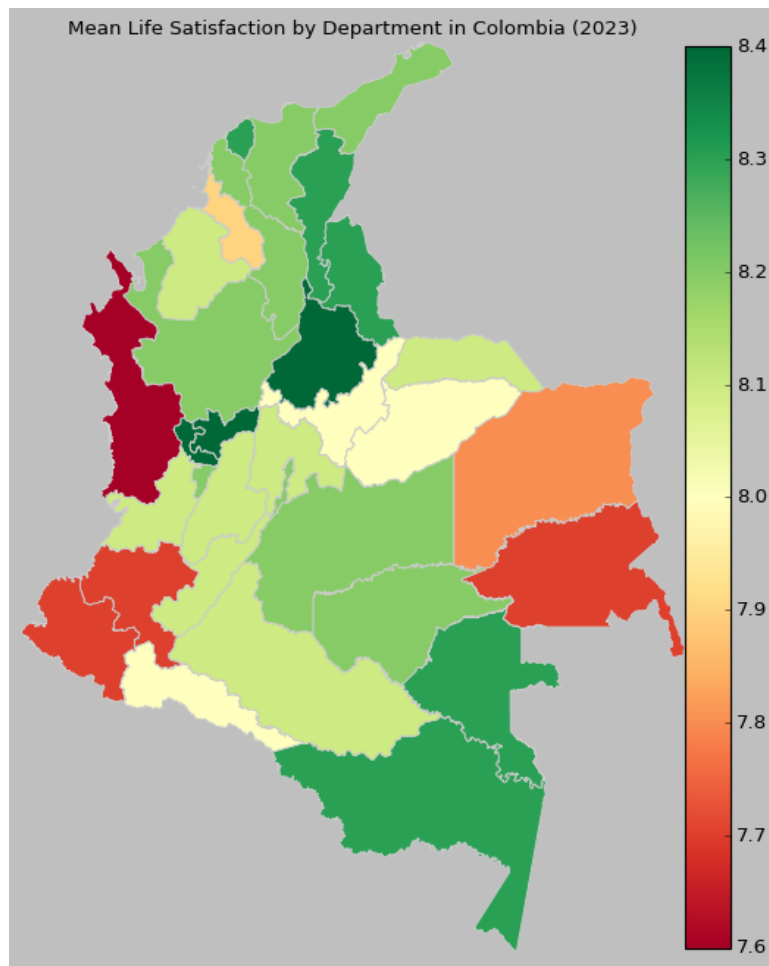
## 2 Departments/Provinces

there are 166k missing out of 242k, we conduct robustness check in appendix using 2021 wave

the map is below, table including number of observations per province is in appendix

unhappy: West (poor Choco, and Southern Cauca and Narino) and East Guainia and Vichada. In 2021 (see appendix) Choco and Cauca were still relatively unhappy, but slightly happier.

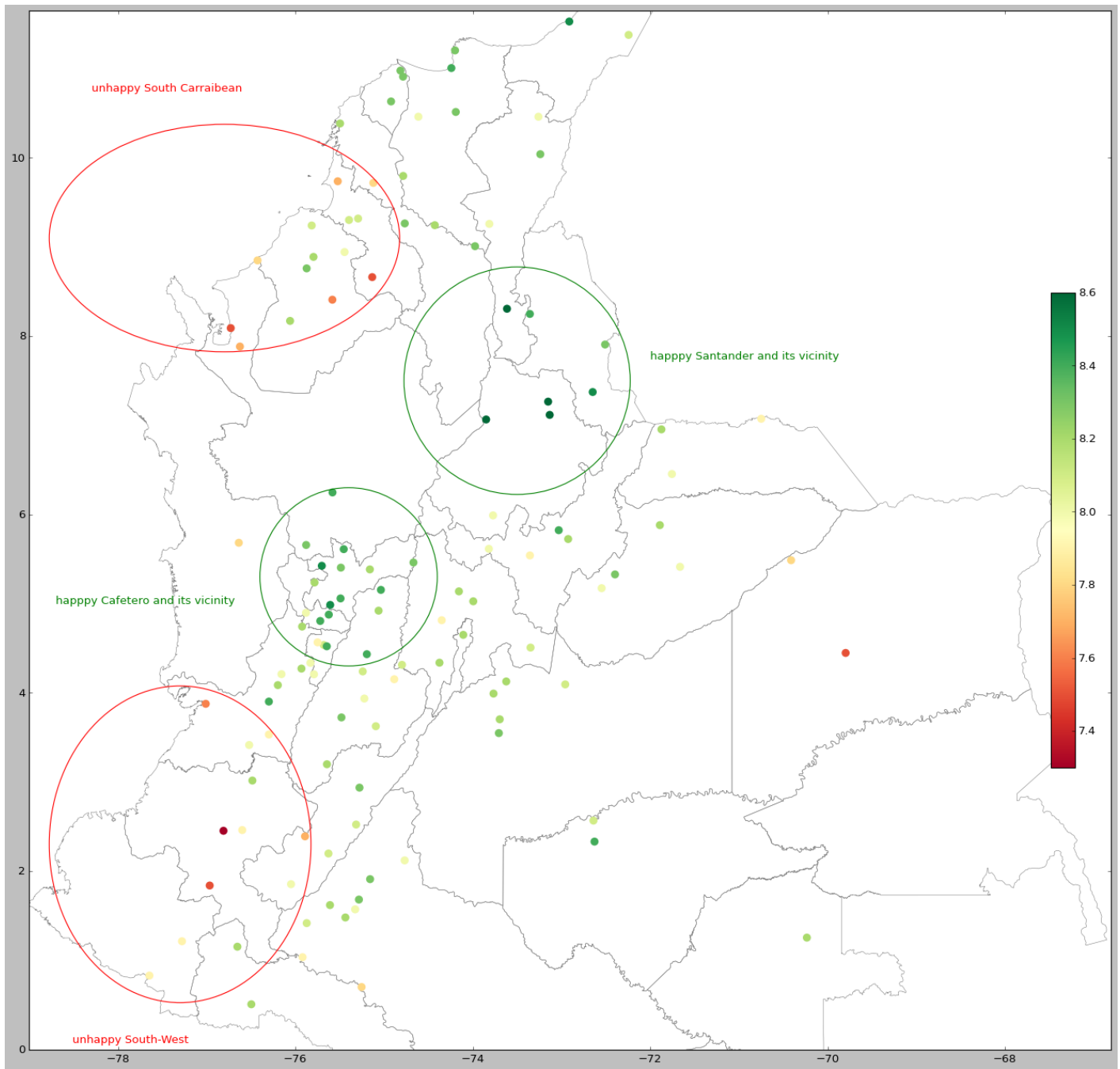
Happy cafetero: Caldas, Risaralda, Quindio—in 2021 there were slight differences (see appendix) but overall Cafetero was happy too. And also happy Santander. Then slightly less happy Norte de Santander, Cesar and Atlantico. In 2021 (see appendix) Santander, Cesar and Atlantico were less happy.



**Figure 1:** Similar to 2021 robustness/sensitivity check in appendix except: small sample Amazonas and Vaupés, both South-Eastern at 8.3 in 2023 v 7.6 in 2021, a .7 difference. Sencond largest differences: Santander and Putumayo increased by .4 from 2021 to 2023.

### 3 Cities/Municipalities

166k/242k missing, again robustness/sensitivity check in appendix with 2021 ecv



**Figure 2:** Zoom in to see the detail

Santander is one of the happiest departments, and so 2 santander largest cities shine: Bucaramanga and Barrancabermeja at 8.6. Bucaramanga was less happy in 2021 at 8.1. Another happy region cafetero has many happy cities: Calarcá, Pereira, Manizales, and Armenia (Armenia was even happier in 2021). Other happy cities include 2nd largest Medellin, Ibaguë, and Ciénaga (Ciénaga was less happy in 2021 at 7.9). The least happy is remote Western Puerto Carreño at 7.5, and several cities at 7.9: Tunja, and Southern cities: Palmira in Valle de Cauca, Popayan in Cauca, and Pasto in Narino.

## 4 Urban-Rural and Farmers

While most developed countries cities are unhappy and indeed unhappy city is a common pattern across the world (Okulicz-Kozaryn and Valente 2021), it is not always the case in developing countries.

here boilerplate from happy col unahhpy bog: that cities are growth engines Glaeser (2011) and O'Sullivan (2009) and developing countries like colombia do need economic growth hence role for cities is gretaer than in already develpd countries

and there is paradox of happy peasant and unhappy millionaire Graham (2009)

though not so much in colombia, urban and rural are about same happy, except by income: bar chart by income category bu urb  
rur as in raman

Both rural and farmer in figures ?? ??–there are slight differences, but overall similar pattern–no difference for the bottom and 9th deciles; rural and farmer happier in 2nd-8th deciles, and urban happier in the top decile. This makes sense, one needs money to enjoy city, where costof living is higher. In addition, life is more commodified in the city–one needs to pay for more things. For instance, many in the rural areas grow their own food. The bottom decile parity could be explained as that there is some minimum income necessary, and below it, it doesnt help to live in rural v urban area.

Overall, it is worth emphasizing that except the 1st decile and up to median and one decile above it–farmers/rural clearly happier.

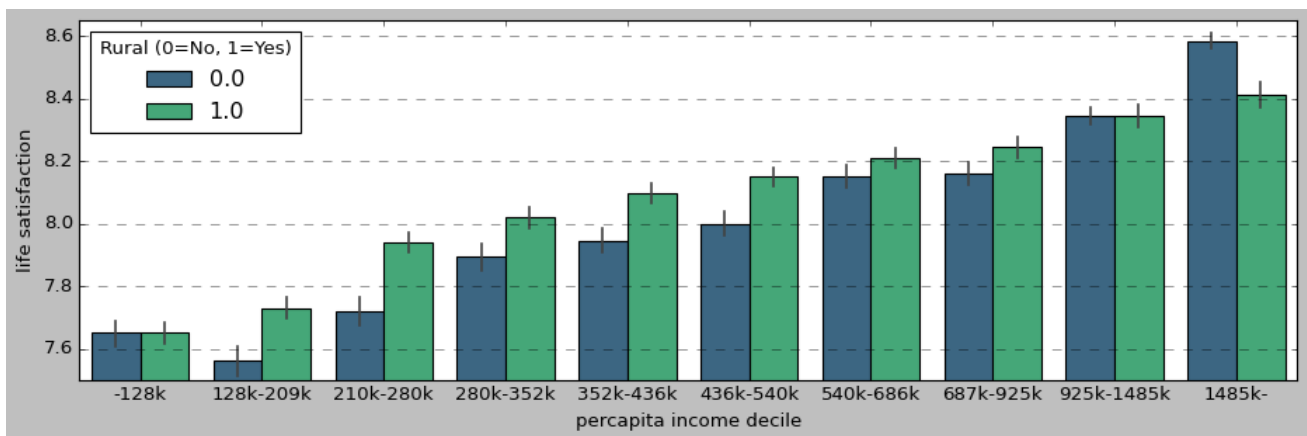


Figure 3

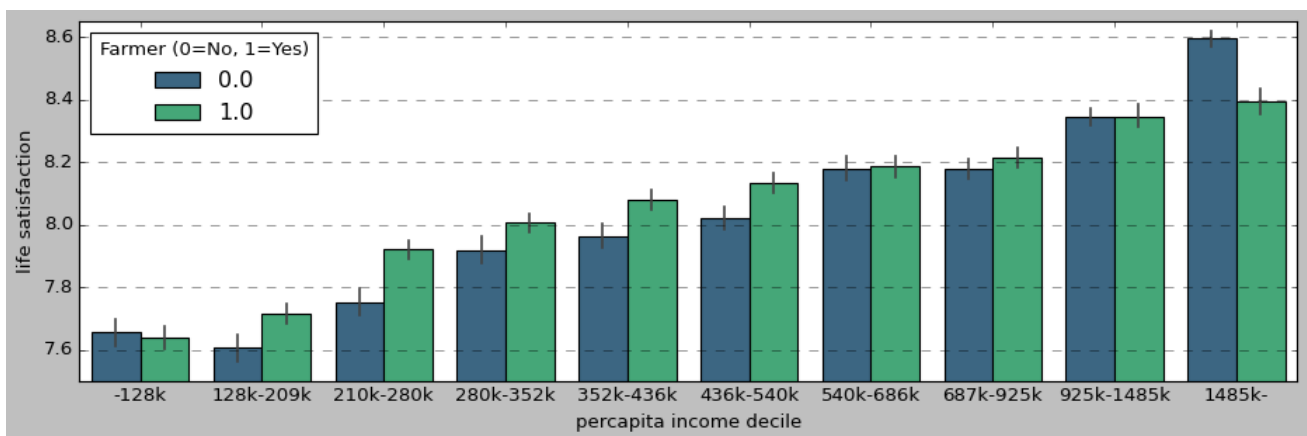


Figure 4

# ONLINE APPENDIX (Supplementary Online Material (SOM))

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

```
. tabstat ls, by(lwd7) stat(mean sd) format(%9.1f)
```

Summary for variables: ls  
Group variable: lwd7 (RECODE of lwd (do what you like))

| lwd7  | Mean | SD  |
|-------|------|-----|
| 0-4   | 5.6  | 2.7 |
| 2     | 7.6  | 2.3 |
| 3     | 7.6  | 1.8 |
| 4     | 8.0  | 1.5 |
| 5     | 7.8  | 1.9 |
| 6     | 8.2  | 1.5 |
| 7     | 8.9  | 1.5 |
| Total | 8.3  | 1.9 |

.9

## 5 Reestimation Using 2023 ECV

2021 May be biased say bc of covid, especially that it peaked in colombia in 2021 <https://coronavirus.jhu.edu/region/colombia>(by 2023 used in the paper it pretty much cleared). But it a useful robustness check, if covid didnt change much , then it should be pretty stable

and it was quite high mortality in col <https://coronavirus.jhu.edu/data/mortality>

### 5.1 Departments/Provinces

Again as said in the paper, overall results are similar to 2023 reported in the body of the paper, except both small sample Amazonas and Vaupés, South-Eastern and with 8.3 in 2023 v 7.6 in 2021, a .7 difference. Sencond largest differences are for Santander and Putumayo which increased by .4 from 2021 to 2023.

**Table 2:** Life satisfaction by department.

| department               | life<br>isfaction<br>2023 | sat-<br>total<br>2023 | obs  | non-<br>missing life<br>satisfaction<br>2023 | mean life satisfaction 2021 | total<br>2021 | obs  | non<br>-ing<br>life<br>satisfaction<br>2021 | 2023-2021 |
|--------------------------|---------------------------|-----------------------|------|----------------------------------------------|-----------------------------|---------------|------|---------------------------------------------|-----------|
| Amazonas                 | 8.3                       | 276                   | 245  | 7.6                                          |                             | 423           | 365  | 0.7                                         |           |
| Antioquia                | 8.2                       | 5499                  | 4729 | 8.3                                          |                             | 6036          | 5060 | -0.1                                        |           |
| Arauca                   | 8.1                       | 974                   | 773  | 8.1                                          |                             | 1103          | 868  | 0                                           |           |
| Atlántico                | 8.3                       | 3144                  | 2311 | 8.0                                          |                             | 2338          | 1799 | 0.3                                         |           |
| Bogotá, D.C.             | 8.2                       | 4688                  | 3446 | 8.1                                          |                             | 5337          | 3810 | 0.1                                         |           |
| Bolívar                  | 8.2                       | 2584                  | 2258 | 8.1                                          |                             | 2495          | 2156 | 0.1                                         |           |
| Boyacá                   | 8.0                       | 4129                  | 3649 | 7.8                                          |                             | 4638          | 4122 | 0.2                                         |           |
| Caldas                   | 8.4                       | 2822                  | 2555 | 8.2                                          |                             | 3335          | 3005 | 0.2                                         |           |
| Caquetá                  | 8.1                       | 2547                  | 2018 | 7.9                                          |                             | 2846          | 2249 | 0.2                                         |           |
| Casanare                 | 8.0                       | 1749                  | 1370 | 8.1                                          |                             | 2051          | 1457 | -0.1                                        |           |
| Cauca                    | 7.7                       | 2580                  | 2244 | 7.9                                          |                             | 2572          | 2287 | -0.2                                        |           |
| Cesar                    | 8.3                       | 2224                  | 1717 | 8.0                                          |                             | 1833          | 1510 | 0.3                                         |           |
| Chocó                    | 7.6                       | 857                   | 754  | 7.9                                          |                             | 1016          | 913  | -0.3                                        |           |
| Cundinamarca             | 8.1                       | 3818                  | 3322 | 7.8                                          |                             | 4233          | 3684 | 0.3                                         |           |
| Córdoba                  | 8.1                       | 2706                  | 2240 | 8.0                                          |                             | 2226          | 1933 | 0.1                                         |           |
| Guainía                  | 7.7                       | 327                   | 273  | 7.4                                          |                             | 350           | 274  | 0.3                                         |           |
| Guaviare                 | 8.2                       | 700                   | 457  | 8.0                                          |                             | 621           | 411  | 0.2                                         |           |
| Huila                    | 8.1                       | 2961                  | 2322 | 8.0                                          |                             | 2999          | 2407 | 0.1                                         |           |
| La Guajira               | 8.2                       | 951                   | 750  | 7.9                                          |                             | 988           | 736  | 0.3                                         |           |
| Magdalena                | 8.2                       | 2458                  | 2096 | 8.0                                          |                             | 2110          | 1710 | 0.2                                         |           |
| Meta                     | 8.2                       | 3683                  | 2752 | 8.1                                          |                             | 3862          | 2817 | 0.1                                         |           |
| Nariño                   | 7.7                       | 2284                  | 2034 | 7.4                                          |                             | 2847          | 2492 | 0.3                                         |           |
| Norte de Santander       | 8.3                       | 2499                  | 2028 | 8.2                                          |                             | 2587          | 2171 | 0.1                                         |           |
| Putumayo                 | 8.0                       | 950                   | 752  | 7.6                                          |                             | 1272          | 1012 | 0.4                                         |           |
| Quindío                  | 8.2                       | 2085                  | 1658 | 8.4                                          |                             | 2186          | 1785 | -0.2                                        |           |
| Risaralda                | 8.4                       | 1634                  | 1353 | 8.3                                          |                             | 1996          | 1663 | 0.1                                         |           |
| San Andrés y Providencia | 8.4                       | 28                    | 25   | 8.2                                          |                             | 29            | 26   | 0.2                                         |           |
| Santander                | 8.4                       | 3790                  | 3194 | 8.0                                          |                             | 4153          | 3574 | 0.4                                         |           |
| Sucre                    | 7.9                       | 1798                  | 1399 | 7.8                                          |                             | 1899          | 1467 | 0.1                                         |           |
| Tolima                   | 8.1                       | 3564                  | 3215 | 8.0                                          |                             | 4257          | 3770 | 0.1                                         |           |
| Valle del Cauca          | 8.1                       | 4892                  | 4093 | 8.1                                          |                             | 5268          | 4421 | 0                                           |           |
| Vaupés                   | 8.3                       | 265                   | 245  | 7.6                                          |                             | 288           | 267  | 0.7                                         |           |
| Vichada                  | 7.8                       | 515                   | 455  | 7.6                                          |                             | 584           | 482  | 0.2                                         |           |

## 5.2 Cities/Municipalities

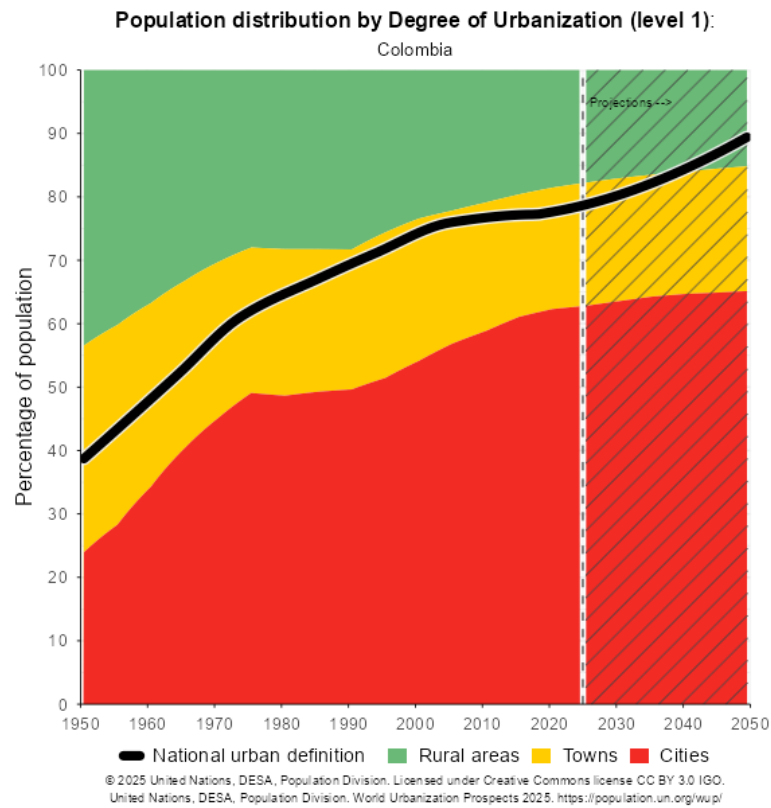
**Table 3:** Life satisfaction by department.

| municipality name      | mean life satisfaction 2023 | non-ing miss-life satisfaction 2023 | mean life satisfaction 2021 | non-ing miss-life satisfaction 2021 | 2023-2021 diff |
|------------------------|-----------------------------|-------------------------------------|-----------------------------|-------------------------------------|----------------|
| Buga                   | NaN                         | NaN                                 | 8.0                         | 215                                 | NaN            |
| Espinal                | NaN                         | NaN                                 | 7.8                         | 220                                 | NaN            |
| Mariquita              | NaN                         | NaN                                 | 7.9                         | 201                                 | NaN            |
| Ocaña                  | NaN                         | NaN                                 | 8.2                         | 200                                 | NaN            |
| San Vicente del Caguán | NaN                         | NaN                                 | 8.1                         | 240                                 | NaN            |
| Armenia                | 8.1                         | 668.0                               | 8.5                         | 681                                 | -0.4           |
| Buenaventura           | 8.1                         | 412.0                               | 8.3                         | 279                                 | -0.2           |
| Cartagena              | 8.2                         | 591.0                               | 8.4                         | 572                                 | -0.2           |
| Facatativá             | 8.1                         | 200.0                               | 8.3                         | 222                                 | -0.2           |
| Palмира                | 7.9                         | 319.0                               | 8.1                         | 332                                 | -0.2           |
| Popayán                | 7.9                         | 225.0                               | 8.1                         | 232                                 | -0.2           |
| Tuluá                  | 8.2                         | 303.0                               | 8.4                         | 307                                 | -0.2           |
| Tunja                  | 7.9                         | 270.0                               | 8.1                         | 241                                 | -0.2           |
| Cali                   | 8.0                         | 985.0                               | 8.1                         | 1108                                | -0.1           |
| Pitalito               | 8.0                         | 293.0                               | 8.1                         | 288                                 | -0.1           |
| Cúcuta                 | 8.3                         | 573.0                               | 8.3                         | 553                                 | 0.0            |
| Manizales              | 8.4                         | 462.0                               | 8.4                         | 567                                 | 0.0            |
| Medellín               | 8.4                         | 905.0                               | 8.4                         | 959                                 | 0.0            |
| Puerto Carreño         | 7.5                         | 225.0                               | 7.5                         | 215                                 | 0.0            |
| San José del Guaviare  | 8.1                         | 246.0                               | 8.1                         | 231                                 | 0.0            |
| Villavicencio          | 8.2                         | 982.0                               | 8.2                         | 990                                 | 0.0            |
| Acacías                | 8.3                         | 229.0                               | 8.2                         | 233                                 | 0.1            |
| Bogotá, D.C.           | 8.2                         | 3446.0                              | 8.1                         | 3836                                | 0.1            |
| Calarcá                | 8.4                         | 244.0                               | 8.3                         | 257                                 | 0.1            |
| Pereira                | 8.4                         | 524.0                               | 8.3                         | 677                                 | 0.1            |
| Santa Marta            | 8.3                         | 225.0                               | 8.2                         | 218                                 | 0.1            |
| Sincelejo              | 8.1                         | 222.0                               | 8.0                         | 275                                 | 0.1            |
| Valledupar             | 8.0                         | 339.0                               | 7.9                         | 274                                 | 0.1            |
| Barranquilla           | 8.3                         | 1363.0                              | 8.1                         | 1116                                | 0.2            |
| Florencia              | 8.2                         | 575.0                               | 8.0                         | 630                                 | 0.2            |
| Ibagué                 | 8.4                         | 478.0                               | 8.2                         | 549                                 | 0.2            |
| Pasto                  | 7.9                         | 244.0                               | 7.7                         | 309                                 | 0.2            |
| Yopal                  | 8.3                         | 264.0                               | 8.1                         | 240                                 | 0.2            |
| Montería               | 8.3                         | 465.0                               | 8.0                         | 372                                 | 0.3            |
| Neiva                  | 8.3                         | 417.0                               | 8.0                         | 436                                 | 0.3            |
| Riosucio               | 8.5                         | 203.0                               | 8.2                         | 203                                 | 0.3            |
| Sogamoso               | 8.2                         | 236.0                               | 7.9                         | 367                                 | 0.3            |
| Aguachica              | 8.6                         | 202.0                               | 8.1                         | 215                                 | 0.5            |
| Bucaramanga            | 8.6                         | 798.0                               | 8.1                         | 842                                 | 0.5            |
| Ciénaga                | 8.4                         | 234.0                               | 7.9                         | 252                                 | 0.5            |
| Magangué               | 8.3                         | 205.0                               | 7.8                         | 207                                 | 0.5            |

## 5.3 Urban-Rural and Farmers

Colombia is quite urbanized at around 70-80%. <https://population.un.org/wup/countryprofiles?country=Colombia&mainCategory=Countries%20and%20Aggregates&subCategory=Urbanization%20-%20International%20definition>

Still, a substantial part at 20-30% is rural.



**Figure 5**

## 5.4 Geography of Colombia

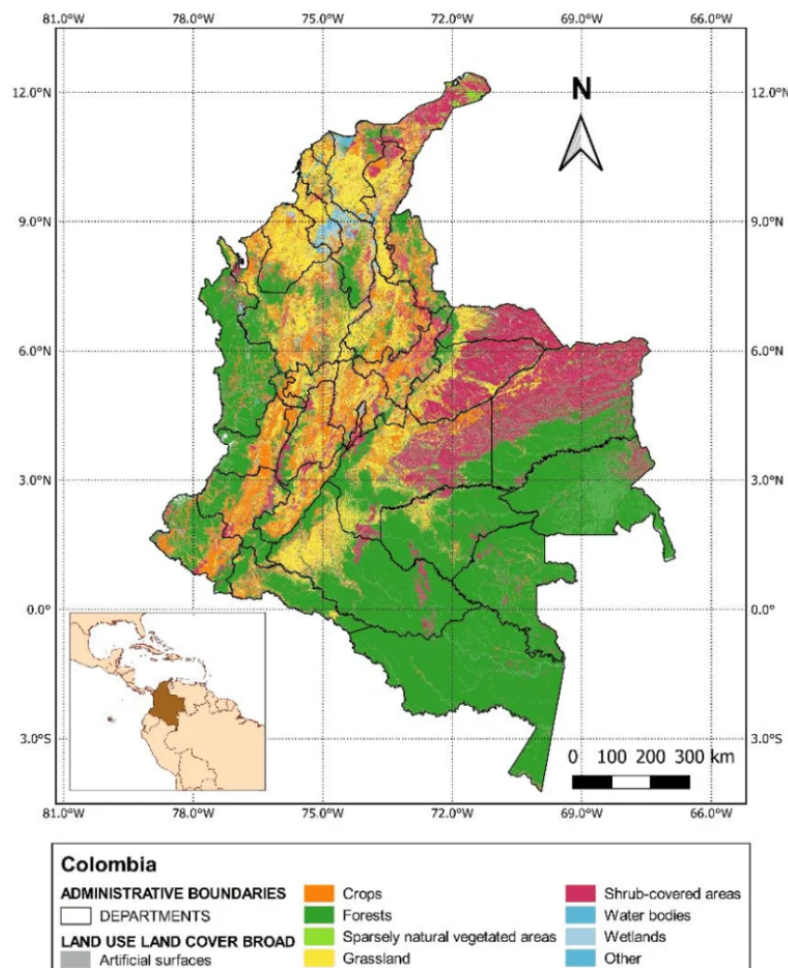
The paper is about geography of happiness, but it would be incomplete without some mention of geography. First map of departments in figure ??.



**Figure 6:** Colombia departments. Source: [https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments\\_of\\_colombia.svg/960px-Departments\\_of\\_colombia.svg.png](https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments_of_colombia.svg/960px-Departments_of_colombia.svg.png).

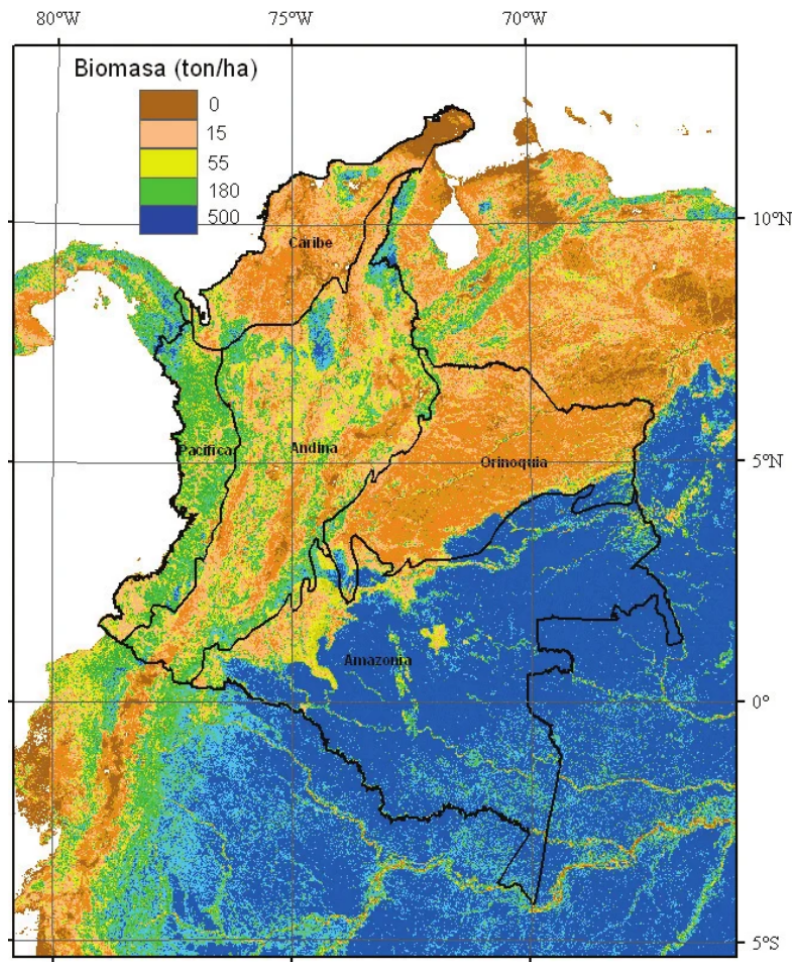
Topography is well mapped and best viewed at [https://en.wikipedia.org/wiki/File:Mapa\\_de\\_Colombia\\_\(relieve\).svg](https://en.wikipedia.org/wiki/File:Mapa_de_Colombia_(relieve).svg), where you can zoom in to see the detail.

Land use is shown in ??—note that most of the country is still natural.



**Figure 7:** Land use. Source: [https://www.researchgate.net/figure/Land-use-and-land-cover-map-of-Colombia-2012\\_fig3\\_346648643](https://www.researchgate.net/figure/Land-use-and-land-cover-map-of-Colombia-2012_fig3_346648643).

Likewise there is a lot of biomass in natural areas (and non-desert and not frozen) in figure ??.



**Figure 8:** Biomass. Source: Anaya, J. A., Chuvieco, E., & Palacios, A. (2008). Estimación de biomasa aérea en Colombia a partir de imágenes MODIS. *Revista de Teledetección*, 30, 5-22. [https://www.researchgate.net/figure/Figura-7-Distribucion-espacial-de-valores-promedio-de-biomasa-en-Colombia\\_fig4\\_255626661](https://www.researchgate.net/figure/Figura-7-Distribucion-espacial-de-valores-promedio-de-biomasa-en-Colombia_fig4_255626661).

Last but not least ecological footprint. <https://data.footprintnetwork.org/> is a useful global overview, Colombian footprint per person is quite low at 2gha, 4x lower than that of the US at 8 gha, and it is stable since 1960s, but biocapcity per person has declined from 11.4 gha in 1960 to 3.3 in 2024. <https://happyplanetindex.org/countries/COL/> is another tool for cross-country comparisons.

<https://www.sciencedirect.com/science/article/pii/S1470160X20305677> provides useful hi-resolution visualizations of footprint across Colombia 1970-2015.

OECD provides more recent update across Colombia: [https://www.oecd.org/en/publications/oecd-environmental-performance-968398f7-en/full-report/biodiversity-conservation-and-sustainable-use\\_2099ac2c.html](https://www.oecd.org/en/publications/oecd-environmental-performance-968398f7-en/full-report/biodiversity-conservation-and-sustainable-use_2099ac2c.html)

## References

- GLAESER, E. (2011): *Triumph of the City: How Our Greatest Invention Makes Us Richer, Smarter, Greener, Healthier, and Happier*, Penguin Press, New York NY.
- GRAHAM, C. (2009): *Happiness around the world: The paradox of happy peasants and miserable millionaires*, Oxford University Press, New York NY.
- OKULICZ-KOZARYN, A. AND R. R. VALENTE (2021): "Urban unhappiness is common," *Cities*, 103368.

O'SULLIVAN, A. (2009): *Urban economics*, McGraw-Hill.

PEIRÓ-PALOMINO, J., A. J. PICAZO-TADEO, AND E. TORTOSA-AUSINA (2020): "The Geography of Well-being in Colombia," Tech. rep.

PNUD (2023): *Informe sobre Desarrollo Humano para Colombia, cuaderno 2. Percepciones y bienestar subjetivo en Colombia: mas alla de indicadores tradicionales.*, <https://www.undp.org/es/colombia/publicaciones/informe-desarrollo-humano-colombia-cuaderno-2>.